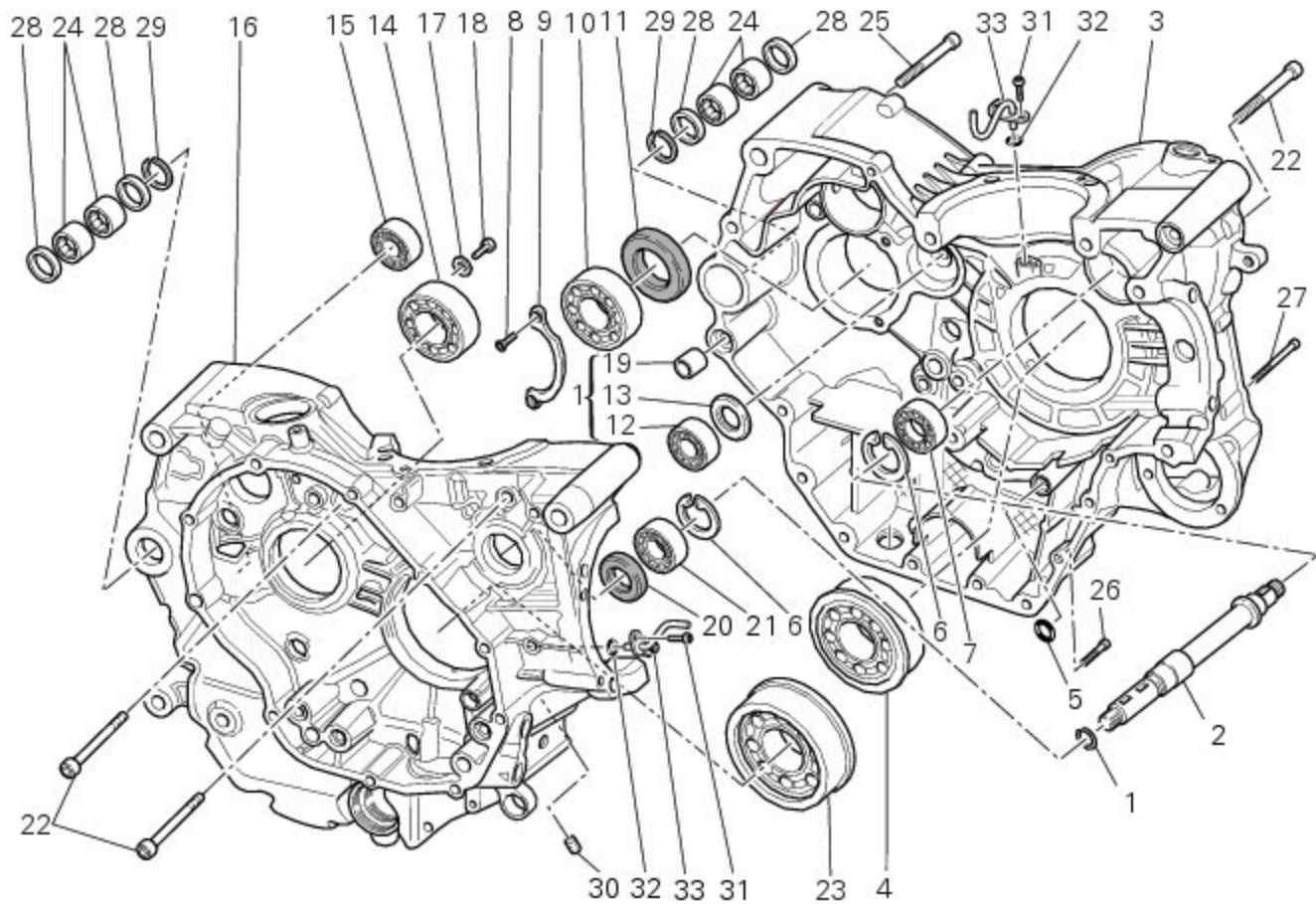


9.2 -Crankcase assembly: crankcase halves



- 1 Circlip
- 2 Timing belt driveshaft
- 3 Alternator crankcase half
- 4 Bearing
- 5 O-ring
- 6 Circlip
- 7 Bearing
- 8 Screw
- 9 Retaining plate
- 10 Bearing
- 11 Sealing ring
- 12 Bearing
- 13 Sealing ring
- 14 Bearing
- 15 Bearing
- 16 Clutch-side crankcase half
- 17 Spacer
- 18 Screw
- 19 Bush
- 20 Sealing ring
- 21 Bearing
- 22 Screw
- 23 Bearing
- 24 Roller bearing
- 25 Screw
- 26 Screw

- 27 Screw
- 28 Sealing ring
- 29 Circlip
- 30 Grub screw
- 31 Screw
- 32 O-ring
- 33 Nozzle

 Spare parts catalogue

- 1100 ABS [TIMING SYSTEM](#)
- 1100 ABS [CRANKCASE HALVES](#)
- 1100 S ABS [TIMING SYSTEM](#)
- 1100 S ABS [CRANKCASE HALVES](#)



Important

Bold reference numbers in this section identify parts not shown in the figures alongside the text, but which can be found in the exploded view diagram.

Separation of the crankcase halves

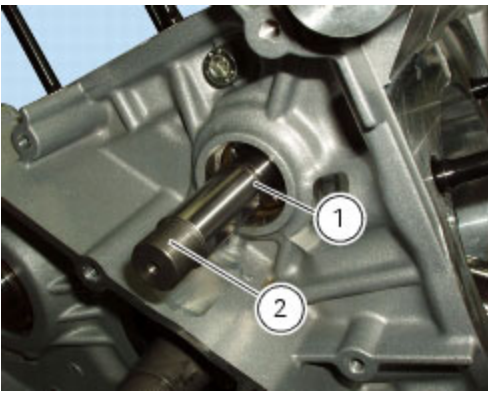
Operation	Section reference
Remove the engine from the frame	N 1, Removal of the engine
Remove the lubrication system	N 2.1, Removal of the oil pump
Remove the timing belts and the timing belt pulleys	N 4.2, Removal of the timing belt covers
Remove the camshafts	N 4.3, Removal of the camshafts
Remove the complete cylinder head unit	N 4.4, Removal of the cylinder head assembly
Remove the cylinder barrel/piston assemblies	N 5, Removal of the cylinder/piston assembly
Remove the alternator-side crankcase cover and the alternator assembly	N 8, Removal of the generator cover
Remove the mesh filter	D 4, Changing the engine oil and filter cartridge
Remove the complete clutch assembly	N 6.1, Removal of the clutch
Remove the clutch-side crankcase cover	N 6.2, Removal of the clutch cover
Remove the primary gear	N 6.3, Removal of the primary drive gear
Remove the starter motor idler gear	N 9.1, Removal of the starter motor idler gear
Remove the starter motor	P 3, Removal of the starter motor

Use two screwdrivers to remove the circlip (1) from the timing belt driveshaft (2) on the clutch-side crankcase half.

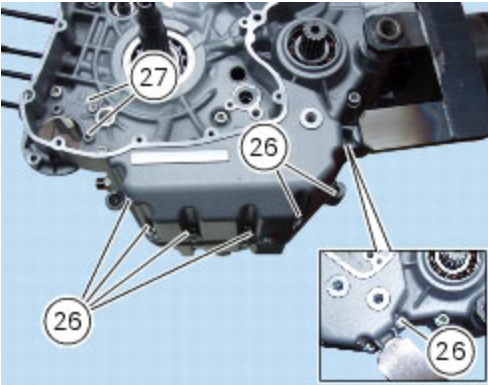
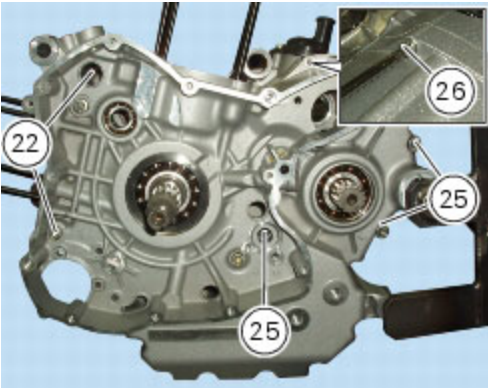


Note

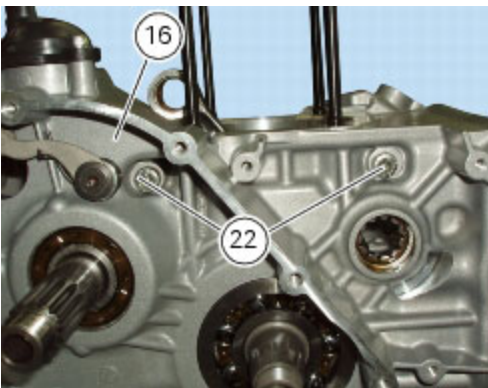
Take care to avoid scoring the surface of the shaft while removing the circlip.



Unscrew the screws (22), (25), (26) and (27) jointing the crankcase halves on alternator side (3).



Unscrew the two screws (22) from clutch-side crankcase half (16), near the vertical cylinder.



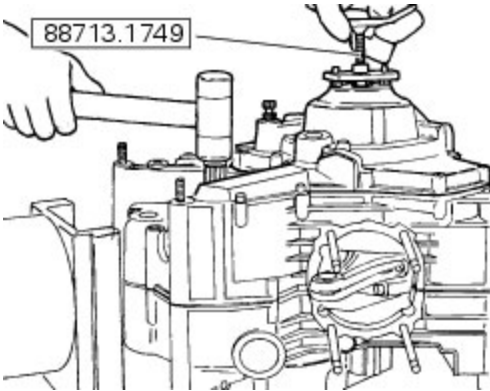
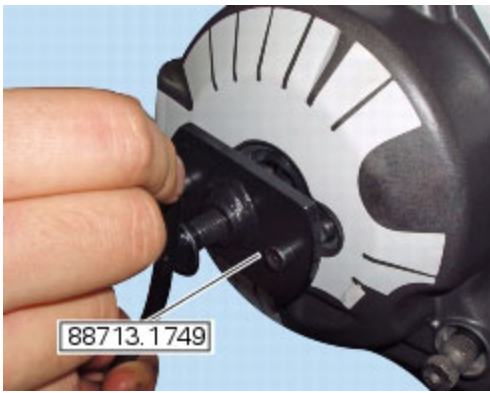
Reuse the generator cover or a service cover with puller **88713.1749** fitted. Secure cover to crankcase half with some of the original screws and begin separation by turning the central pin of the tool.

Tap the end of the gearbox secondary shaft with a plastic mallet to separate the crankcase halves.



Note

Take care not to lose the shims on the shafts and on the selector drum.

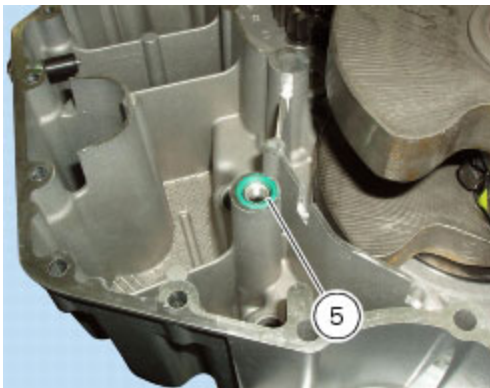
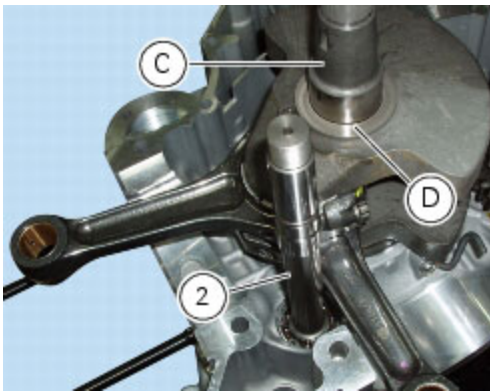


Remove gearbox shafts and gear selector drum from the crankcase halves (Sect. N 7.2, [Removal of the gearbox assembly](#)).

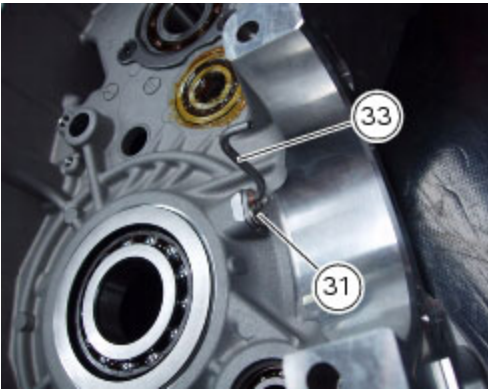
Drive out the crankshaft (C) using a plastic mallet, taking care not to lose the shims (D).

Remove the timing belt driveshaft (2).

Remove the O-ring (5) on the oilway connecting the two crankcase halves.



Remove the screws (31), remove the pipes (33) and collect the O-rings (**32**).



Overhaul of the crankcase halves

Carefully examine the engine crankcase halves.

Check that the surfaces of the crankcases are perfectly flat using a reference surface. Check that the bearings (4) and (23) are in good condition. Note that the main bearings must always be changed in pairs (refer to the procedure below in "[Overhauling the main bearings](#)").



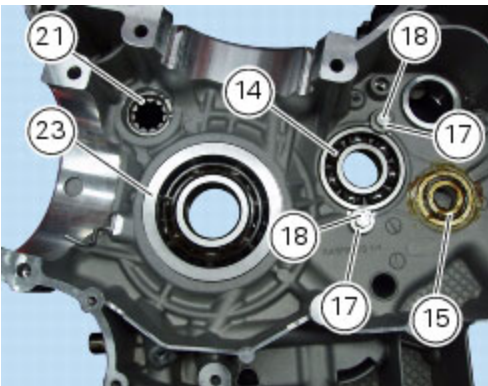
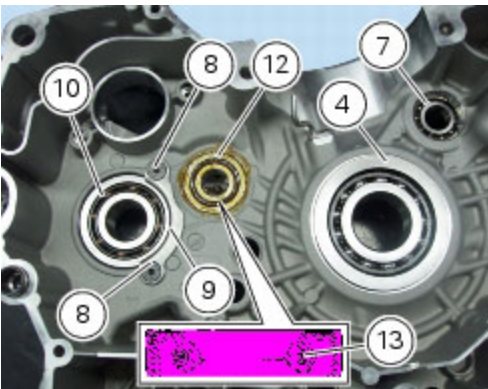
Note

When overhauling an engine it is good practice to renew all the crankcase half bearings.

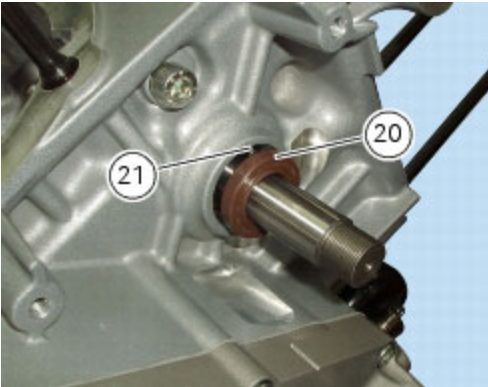
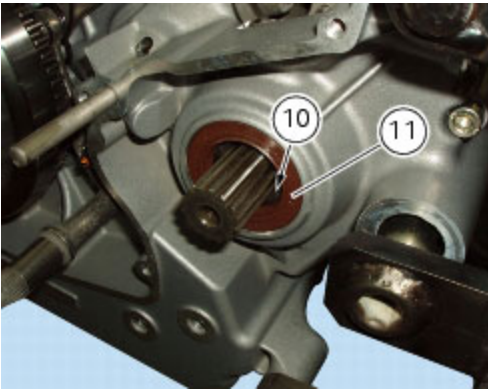
Renew the bearings (10) and (15) on the ends of the gearbox secondary shaft and timing belt driveshaft bearings (7) and (21).

When renewing the primary shaft bearings (14) and (12), install them to the crankcase half using spacers (17) and stop plate (9) and fasten with screws (18) and (8) respectively.

At each overhaul it is recommended to renew also the sealing ring (13) in-between bearing (12) and chain-side crankcase half: lubricate the ring (13) with denatured alcohol, drive it fully home against the crankcase half, and orient it as shown.



At each overhaul it is recommended to renew also the oil sealing ring (11) on the outside of bearing (10) as well as oil sealing ring (20) on the outside of bearing (21).

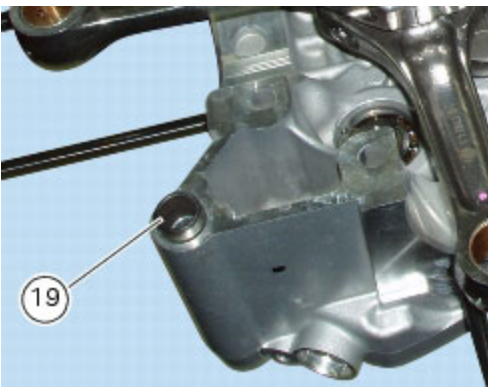
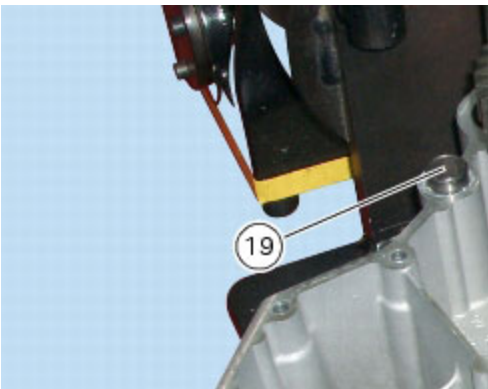


Check the condition of locating bushes (19). If apparently distorted or loose in their seats, change them using proper tools.

When the locating bushes (19) are hard to remove from casing, use a left-hand tap to force bushes out.

Important

The bushes (19) must always be renewed when they have been removed using the above procedure.

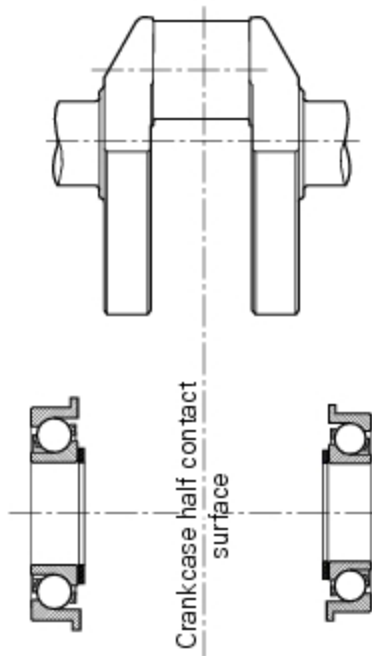


The main bearings have are of the angular contact type with offset inner races so that the balls transmit loads from one groove to the other along straight lines at an angle to the axis of the bearing. The angle-contact ball bearings are designed for bearing combined loading (radial-axial loads).

Bearings of this type can bear thrust loads in one direction only. In fact, under the action of a radial load inside the bearing, an axial force is created that must be counterbalanced by an axial force acting in the opposite direction; that is why these bearings are generally fitted back to back in pairs.

To renew the bearings proceed as follows:

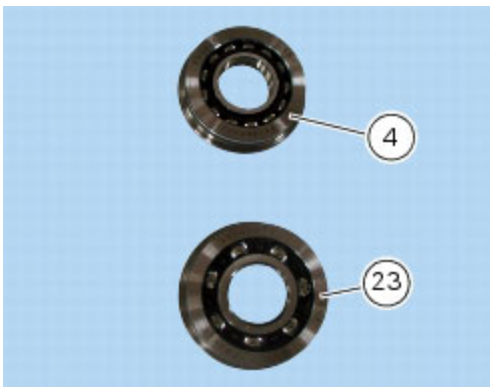
- heat the crankcase half in an oven to **100 °C**;
- remove the bearing using a drift and hammer;
- install the new bearing (while the crankcase is still hot) keeping it perfectly square in its seat using a tubular drift that only bears on the outer ring of the bearing;
- allow the parts to cool and check that the bearing is securely seated in the crankcase half.



Important

On badly worn engines, bearing outer rings may have developed clearance in the crankcase halves - normally, bearing outer rings are interference-fit in the casing.

Check that the interference fit between the crankcase and the bearings (4) and (23) installed, is not less than **0.03 mm**, otherwise, replace the crankcase halves.



Important

After installing new main bearings, shim the crankshaft as described in the paragraph "[Shimming the shafts](#)" and "[Reassembly of the crankcase halves](#)" of this section.

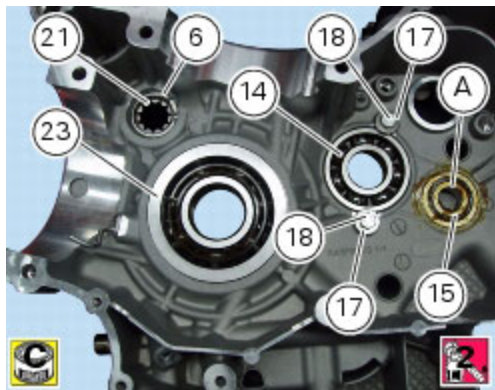
Reassembly of the crankcase halves

The crankcase halves must be in good condition and perfectly clean. The mating surfaces must be perfectly flat and free from burrs.

Refitting the clutch-side crankcase half

The following parts must be present on the internal side of the crankcase half:

- gearbox secondary shaft bearing (15): apply grease to the bearing rollers. Fit the inner race (A) removed previously from the bearing. Apply grease to the inner ring;
- the primary shaft bearing (14), secured with screws (18) and retaining spacer (17): apply prescribed threadlocker to the screws (18) and tighten to the specified torque (Sect. C 3, [Engine torque settings](#));
- the roller bearing (21) with retaining circlip (6) installed in correspondence with the end of the timing belt driveshaft.



Refitting the alternator-side crankcase half

The following parts must be present on the internal side of the crankcase half:

- the double race ball bearing (10) supporting the selector fork shaft.

Apply threadlocker to the screws (8). Fit the retaining plate (9) to the gearbox secondary shaft bearing (10) screwing the screws (8) fully into the chain-side crankcase half.



Note

The bearing retaining plate must be positioned so that the countersunk side is facing upwards.

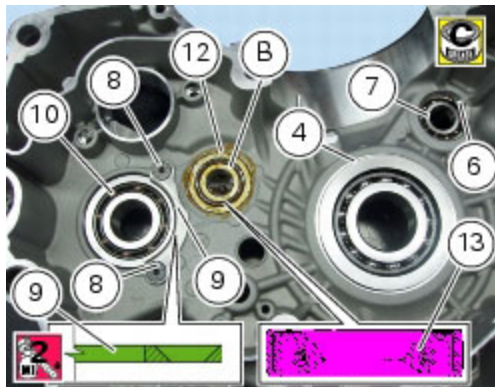
Tighten the screws (8) to the specified torque (Sect. C 3, [Engine torque settings](#)).

The gearbox primary shaft end bearing (12) with the inner spacer (13): apply grease to the bearing rollers.

Fit the inner race (B) removed previously from the bearing. Apply grease to the inner race.

The main bearing (4).

The ball bearing (7) with circlip (6), in correspondence with the timing belt driveshaft, oriented so that the closed side of the plastic roller cage faces the crankcase half.



Note

Bearings do not have a specific mounting position (except main bearings); however it is good practice to install the bearings so that side bearing the writing is facing upwards.

If the swingarm pivot (24) bearings were removed, proceed as follows.

Fit the snap ring (29) into bore (H) of the relevant casing (3) and/or (16), and install it in its seat (L).

Apply some recommended grease inside bore (H).

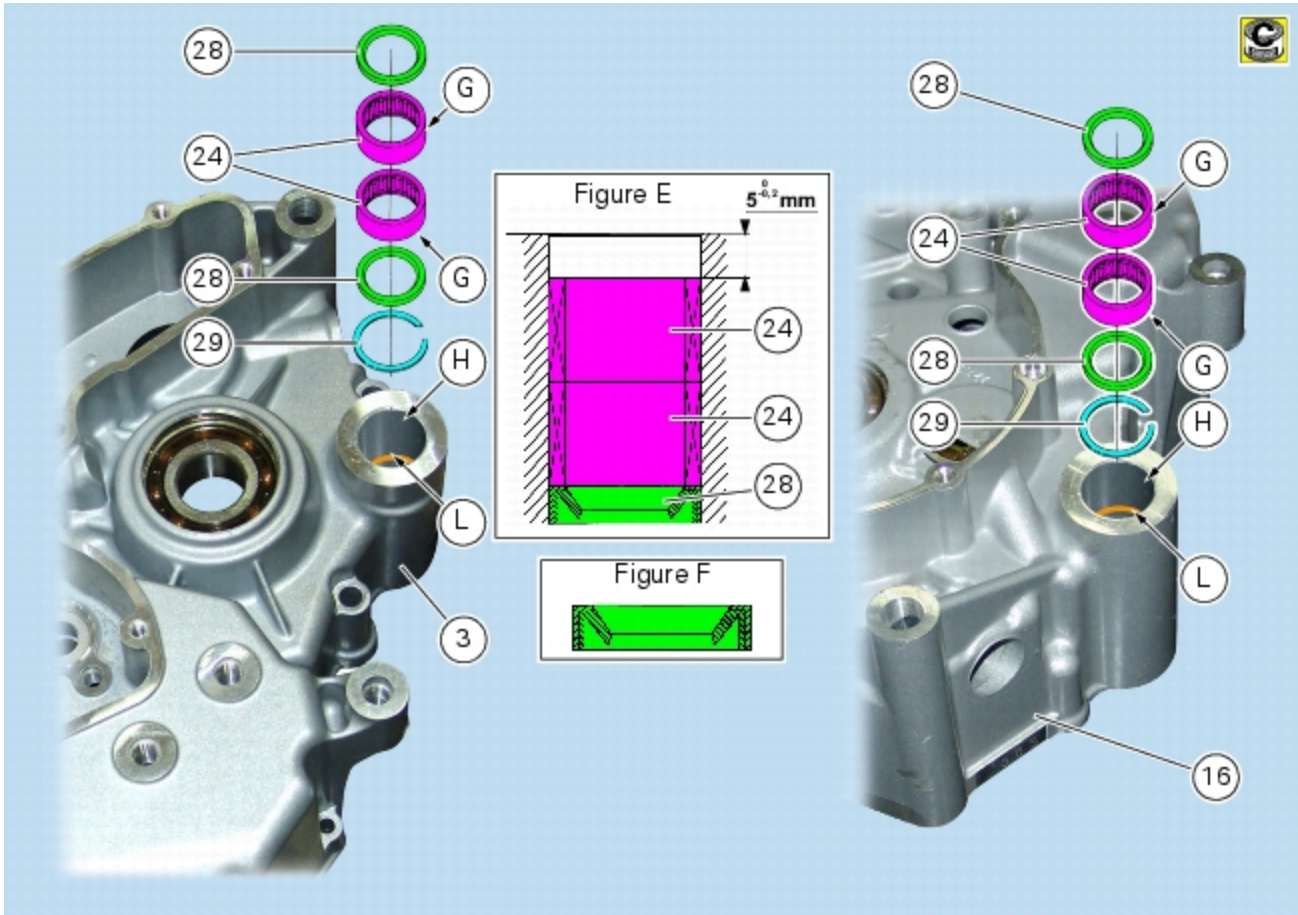
Fit seal (28) and needle roller bearings (24), at the height indicated in figure "E", and ensuring that the three parts are fully home one against the other.

During installation, position the three parts as explained below:

seal (28) shall be positioned as shown in figure "F" and needle roller bearings (24) shall be mounted so that the side with "writing" (G) is facing the casing.

Install seal (28), positioning it as shown in figure "F", fully home against needle roller bearing (24).

Apply some recommended grease inside the needle roller bearings.



Shimming the shafts

Before assembling the crankcase halves, calculate the shims required to obtain the correct end float of the crankshaft and gearbox shafts.

To determine the correct shim thickness proceed as follows.

Shimming the crankshaft

After fitting the new main bearings, proceed as follows to determine the total shim thickness "SA":

measure the distance "LA" between the bearing contact surfaces on the crankshaft;

measure the depths "LA1" and "LA2" corresponding to the distance between contact surface of the crankcase half and the contact surface of the inner race of the bearings.

add a preload of **0.30** mm to help seat the axial load crankshaft bearings correctly, then insert the inner race in the outer ring.

Thus, we obtain:

$$SA = LA1 + LA2 + 0.30 - LA.$$

To calculate the thickness of each shim note that:

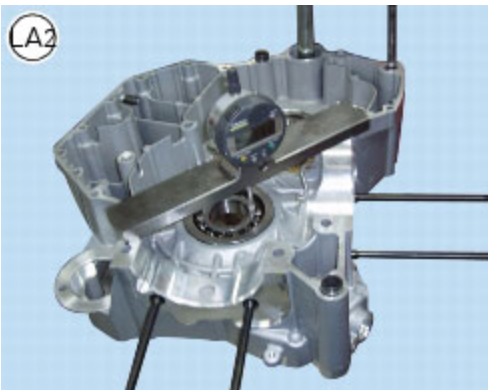
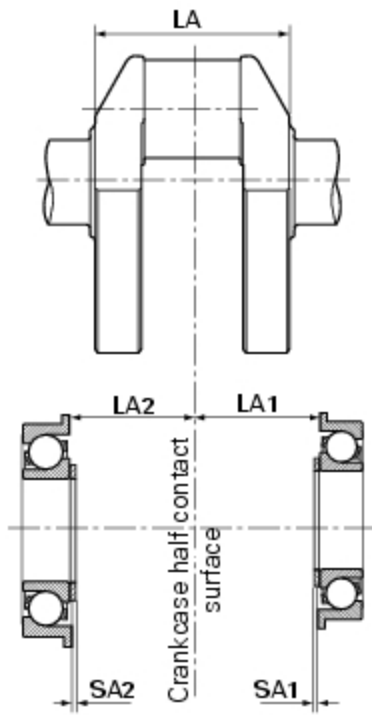
$$SA = SA1 + SA2$$

where "SA1" and "SA2" represent the shims for the clutch-side crankcase half 1 and the alternator-side crankcase half 2.

Considering the alignment of the shaft, this gives:

$$SA1 = LA1 + 0.15 - LA/2;$$

and finally, the second shim thickness: $SA2 = SA - SA1$.



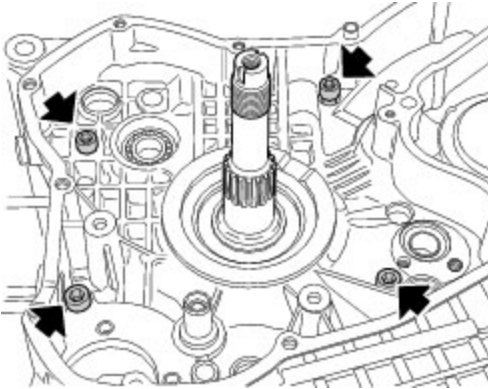
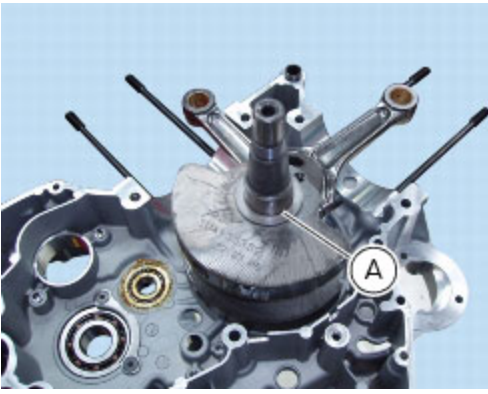
Practical method for determining crankshaft shims

In addition to the above procedure, the following is a practical shimming method, providing a guide on how to calculate the crankshaft shim thickness accurately.

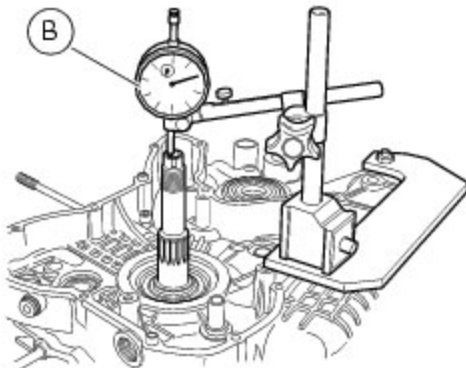
Install a shim (A) of minimum thickness (**1.90 mm**) on each side of the crankshaft to prevent contact between the crankshaft web and the engine block.

Install the crankshaft and assemble the two crankcase halves.

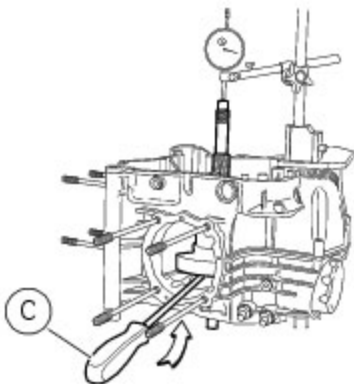
Fit four M8 screws in the seats indicated in the figure and tighten them to the specified torque (Sect. C 3, [Engine torque settings](#)).



Position a dial gauge (B) with magnetic stand on a support plate fixed to the crankcase. Bring the stylus into contact with the end of the crankshaft and set the dial gauge to zero in this position.



Place a lever (C) between crankcase and crank web of the crankshaft pushing towards dial gauge.



Note on the dial gauge the total clearance and add a preload of **(0.30 mm)** plus the thickness of the shims used **(1.90x2=3.8 mm)**.

Divide the resulting value by two to obtain the thickness of the shim packs to be installed at either end of the crankshaft.



Note

After assembling the crankcase halves, the crankshaft should turn with some interference in the new bearings.

Shimming the gearbox shafts

The following thickness spacers are supplied as spare parts.

Position	Clutch side (mm)	Chain side (mm)
Gearbox primary shaft	1.2	1.6
Gearbox secondary shaft	3.15	0.8

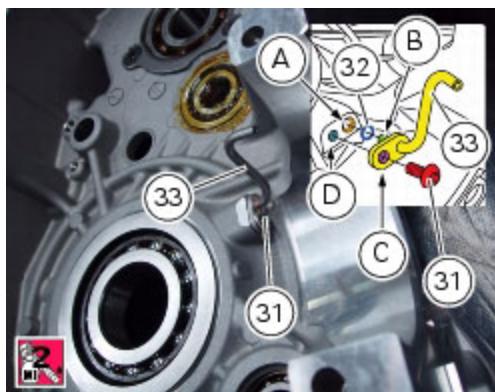
Shimming the gearbox selector drum

The following thickness spacers are supplied as spare parts.

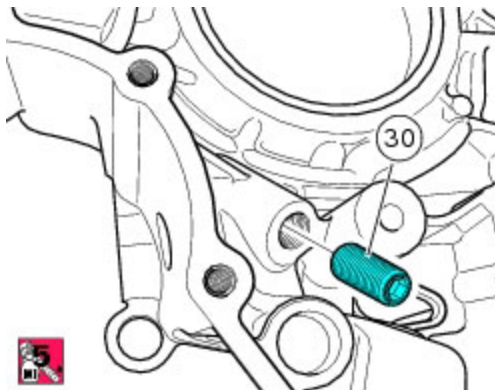
Position	Clutch side (mm)	Chain side (mm)
Gear control drum	1.0	0.9

Reassembly of the crankcase halves

Install the O-ring (32) on its seat (A) in the crankcase half. Fit the pipe (33) in the crankcase half, inserting the end (B) fully in the O-ring. Align the hole (C) of the pipe with the threaded hole (D) in the crankcase half. Apply threadlocker to the screw (31) and start it in the threaded hole (D). Tighten the screw to the specified torque (Sect. C 3, [Engine torque settings](#)).



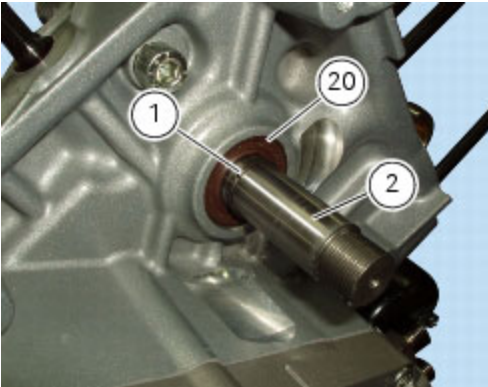
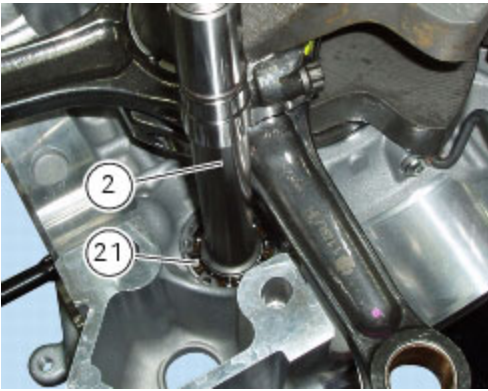
If removed, apply recommended threadlocker on the dowel (30) and tighten to the specified torque (Sect. C 3, [Engine torque settings](#)).



Install the timing belt driveshaft (2) in the roller bearing (21) in the clutch-side crankcase half.

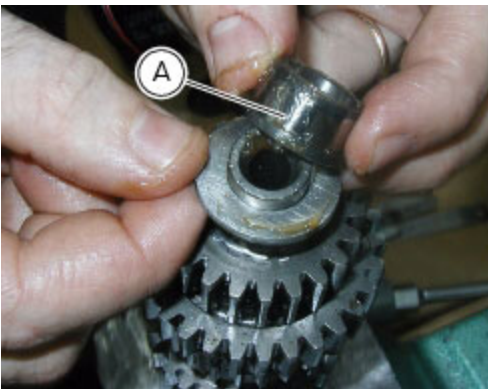
To avoid damaging the oil sealing ring (20) on the timing belt driveshaft, protect the threaded end of the shaft with the special protective cap **88700.5749**. Dampen the oil sealing ring (20) with alcohol and fit it on the timing belt driveshaft, seating it against the roller bearing (21).

Fit the circlip (1) in the groove on the shaft and remove the protective cap.

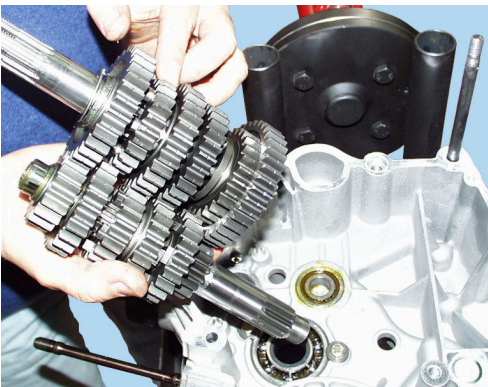


Note

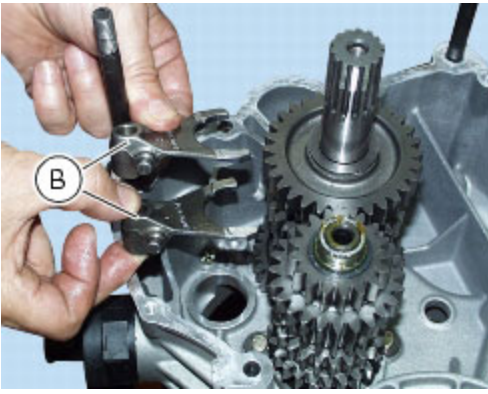
When refitting used components, before installing the gearbox assembly in the crankcase half, make sure that the inner races (A) of the gearbox shaft bearings are fitted into the correct bearings and have not been left on the shafts.



Mesh the gearbox shafts and install them on the clutch-side crankcase half with the calculated shims.



Insert the 1st-4th and 2nd-3rd speed selector forks (B) in their respective channels in the driven gears of the secondary shaft.



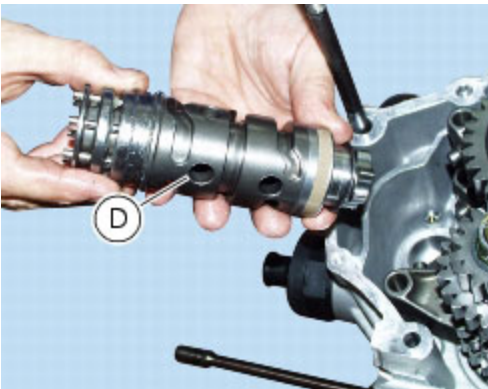
Note

The two selector forks are identical.

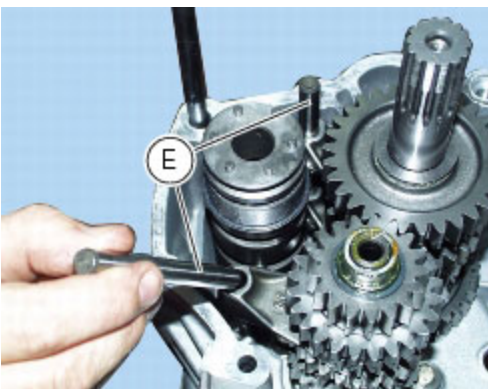
Insert the 5th-6th speed selector fork (C) in the driven gear of the primary shaft.



Install the selector drum (D) in the crankcase with calculated shim thickness.



Fit the pins (E) in the previously installed selector forks.

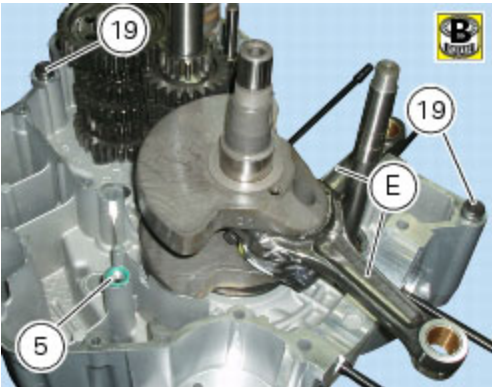


Fit the shimmed crankshaft into the clutch-side crankcase half, positioning the connecting rods (E) in correspondence with the relative cylinder barrel seats.

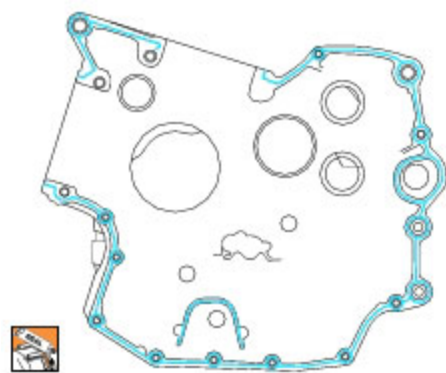
Check that the two locating bushes (19) are correctly fitted.
Grease the O-ring (5) to hold it in position and install it on the oil way connecting the two crankcase halves.

 Important

Make sure that the connecting rods (E) are correctly positioned in the respective cylinder seats. Incorrect positioning of the connecting rods at this stage will inevitably lead to the need to re-open the crankcase half.



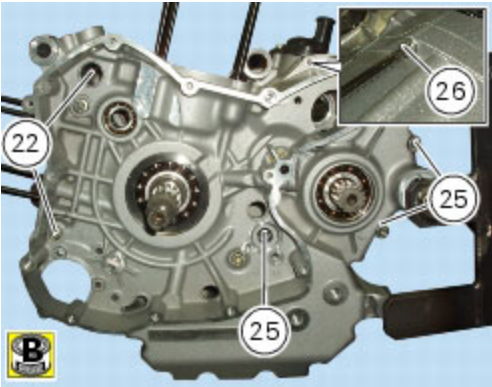
Apply a uniform and continuous bead of DUCATI liquid gasket to the mating surfaces of the crankcase halves, going around all the holes as shown in the figure.

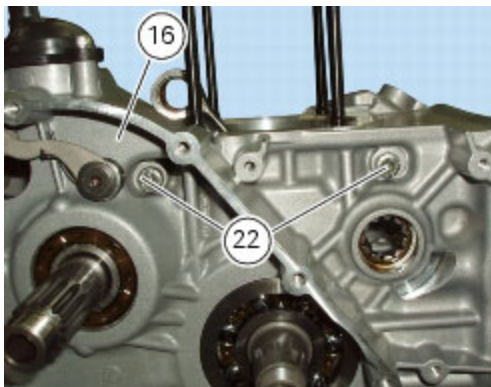
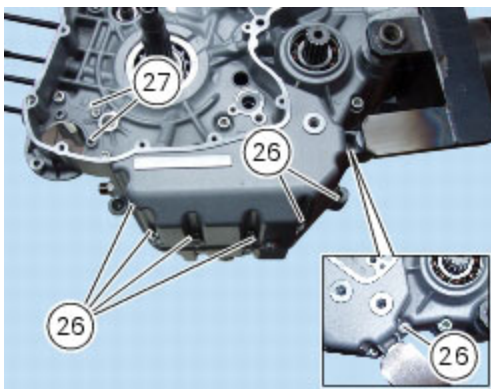


Bring the crankcase halves (3) and (16) together, tapping with a rubber mallet in the area of the shafts if necessary. Prepare the crankcase screws and insert them in the alternator-side crankcase half (3); note that the screws are of different lengths.

Ref.	Qty	Description
22	4	M8x90 mm screws
25	3	M8x75 mm screws
26	8	M6x35 mm screws
27	2	M6x75 mm screws

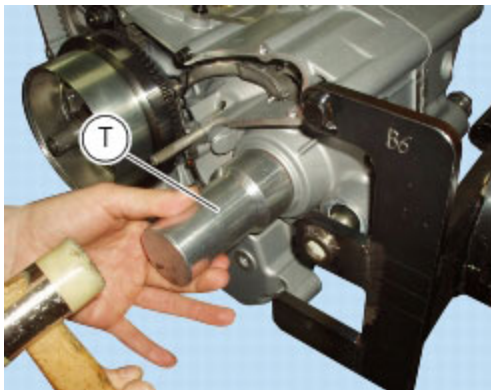
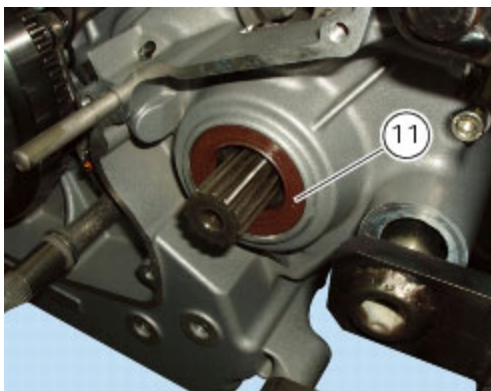
Progressively tighten the crankcase screws until fully seated, starting with the screws of the largest diameter (M8). Insert two M8 screws (22) in the clutch-side crankcase half (16). Tighten all the screws to the specified torque (Sect. C 3, [Engine torque settings](#)).





Check that the crankshaft can be turned with a certain amount of interference with the main bearings (the crankshaft should have a pre-load of **0.15 to 0.20 mm**); check also that all the parts you have fitted are free to rotate or move correctly.

Proceed to install the oil sealing ring (11) and the spacer with the O-ring in the gearbox output shaft using the specific tool (T).



Operation	Section reference
Refit the starter motor	P 3, Refitting the starter motor

Refit the starter motor idler gear	N 9.1, Refitting the starter motor gear
Refit the primary gear	N 6.3, Fitting the primary drive gear pair and checking backlash
Refit the clutch-side crankcase cover	N 6.2, Refitting the clutch cover
Refit the clutch assembly	N 6.1, Reassembling the clutch
Refit the mesh filter	D 4, Changing the engine oil and filter cartridge
Refit the alternator-side crankcase cover and alternator assembly	N 8, Refitting the generator cover
Refit the cylinder barrel/piston assemblies	N 5, Refitting the cylinder/piston assembly
Refit the complete cylinder head unit	N 4.4, Refitting the cylinder head assembly
Refit the camshafts	N 4.3, Refitting the camshafts
Refit the timing belts and the timing belt pulleys	N 4.2, Refitting the timing belt covers
Refit the lubrication system	N 2.1, Refitting the oil pump
Refit the engine to the frame	N 1, Refitting the engine